

KSRM UMS: Enterprise Architecture

Next-Generation Cloud-Native University Operating System. Scalable. Secure.
Autonomous.

01. The Foundation

Architecting for Speed, Stability, and Absolute Cost Optimization.

Core Technology Infrastructure



Next.js 16

Leveraging Server Components and Edge Streaming for sub-100ms initial load times and absolute SEO performance.



Supabase & Prisma

PostgreSQL with Row-Level Security (RLS) and Type-safe ORM ensuring data integrity and massive scale.



Vercel Serverless

Autoscaling infrastructure that spins up thousands of Lambda functions instantly during Result Day peaks.

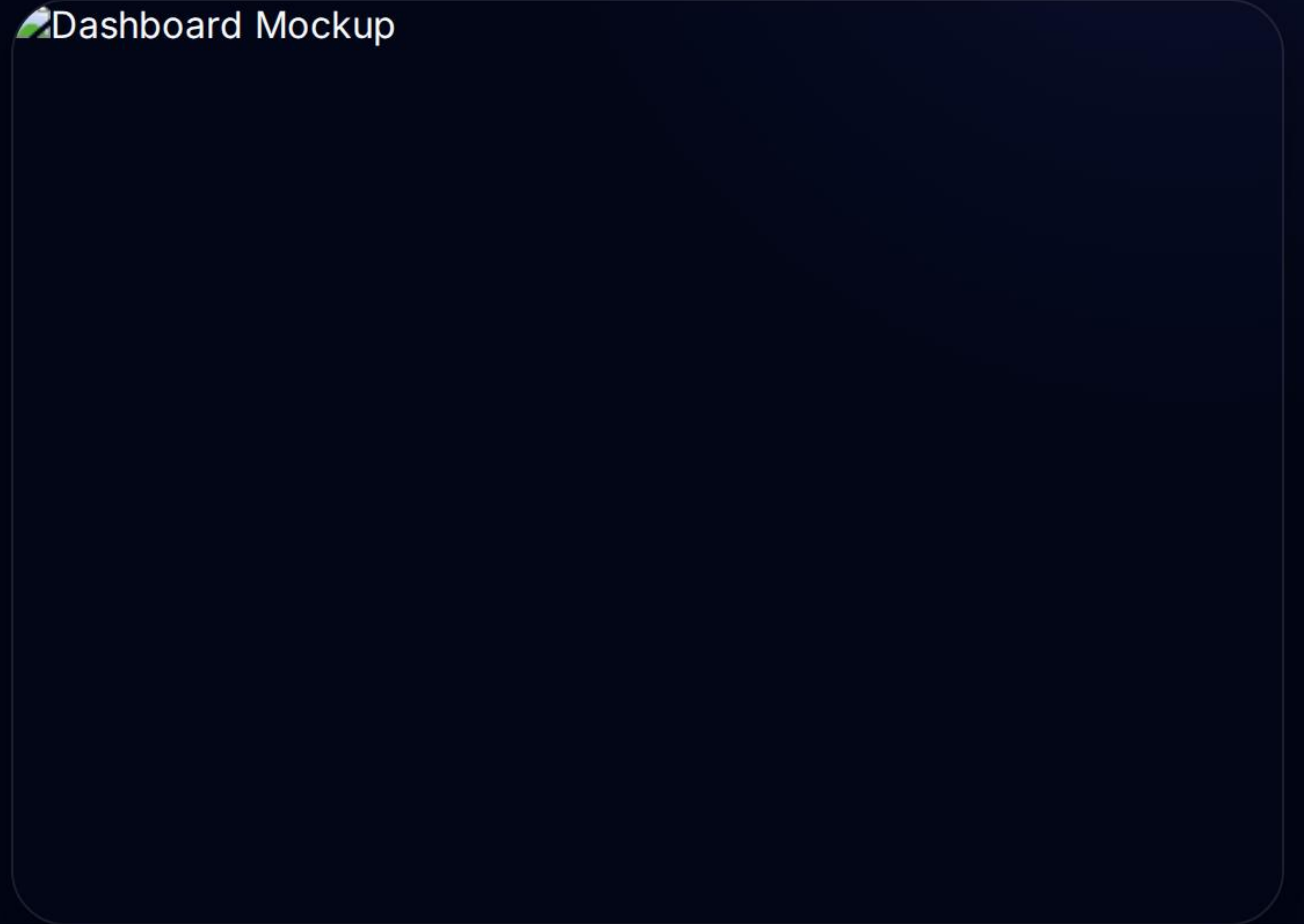
| Visual Intelligence Dashboard

Data-Driven Governance

Our centralized command center provides administrators with real-time visibility into institutional health.

- ✓ **Unified View:** Real-time attendance, finance, and disciplinary metrics on a single pane.
- ✓ **Smart Alerts:** Autonomous notifications for fee defaulters and critical academic drops.

 Dashboard Mockup



| Performance & Financial Impact

99.9%

UPTIME RELIABILITY

Eliminating Operational Chaos

Legacy systems crash under load. Our serverless architecture distributes traffic across the global edge network, guaranteeing stability during high-stakes events like exams and result declarations.

Revenue Capture: Algorithmic enforcement ensures zero fee leakage by automating hall-ticket blocking for defaulters.

| Operational Cost Optimization

Traditional (On-Prem)

₹40L/Year



Antigravity UMS

₹6.5L/Year

Our Serverless model eliminates the need for physical server rooms, 24/7 AC cooling, and dedicated on-site IT hardware maintenance teams.

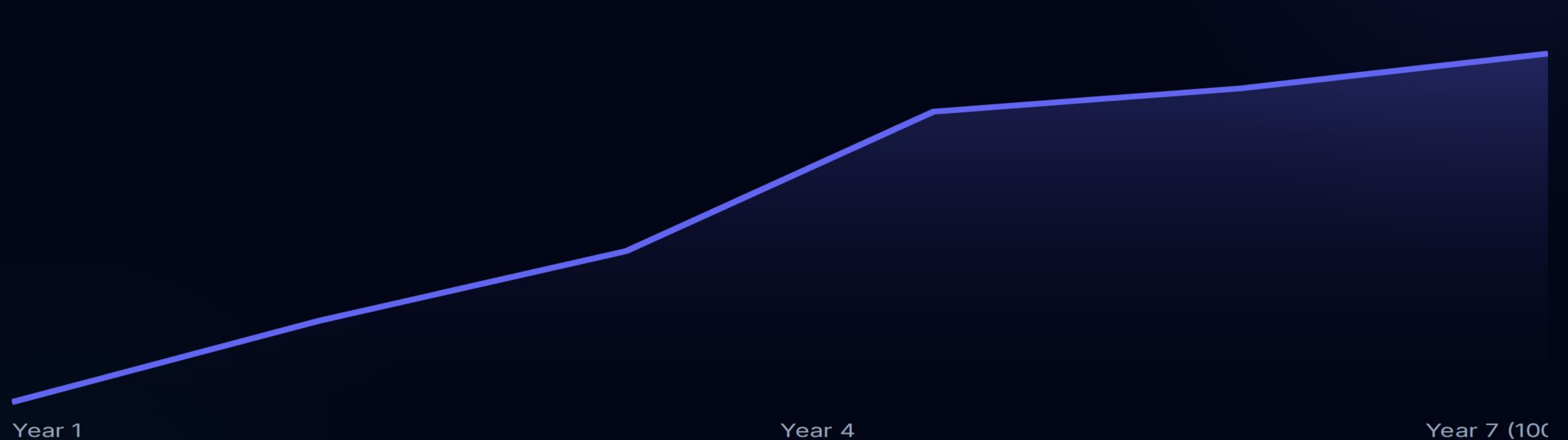
| Current Feature Capabilities

Module	Core Capabilities	Status
Security Vault	IP-based Audit Logging, Role-Based Access Control (RBAC).	LIVE
Financial Ledger	Autonomous Fee Reconciliation, Receipt Automation.	LIVE
Academic Hub	Mark Registration, SGPA/CGPA Calculation, Exam Scheduler.	LIVE

02. Strategic Scaling

From a Single College to a Statewide Educational SaaS Network.

| System Scaling & Adoption Projection



Architecture built for 100,000+ concurrent users through Multi-Tenant Row-Level Security.

AI Curriculum Intelligence

RAG Integration: Our system ingests thousands of JNTU syllabus documents into a centralized vector database.

Administrators and students can ask complex academic queries and receive accurate, citation-backed answers directly from the official curriculum vault.

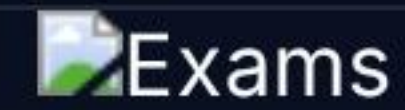


| Upcoming Ecosystem Modules



Auto-Scaling SaaS

Onboarding 100 institutions via automated subdomain provisioning.



Parameterized Exams

Algorithmic question generation to prevent paper leaks and cheating.



Schedule Optimizer

Constraint-based AI solvers for dynamic college timetable generation.

Questions?

Thank you for your vision and partnership.

 antigravity.ai |  ceo@antigravity.ai

Image Sources



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